

- (a) 34 years (b) 36 years
(c) 40 years (d) 38 years
(e) 28 years

55. Rahul is as much younger than Sagar as he is older than Purav. If the sum of the ages of Purav and Sagar is 66 years, and Sagar's age is 48 years, then what is Purav's age? (in years)

[NICL—AAO Exam, 2015]

- (a) 18 (b) cannot be determined
(c) 16 (d) 20
(e) 12

56. 4 years ago, the ratio of $\frac{1}{2}$ of A's age at that time and four times of B's age at the time was 5 : 12. Eight years hence, $\frac{1}{2}$ of A's age at that time will

be less than B's age at that time by 2 years. What is B's present age?

[IBPS—RRB Officers Exam, 2015]

- (a) 10 years (b) 14 years
(c) 12 years (d) 5 years
(e) 8 years

57. Ten years hence, the respective ratio between Simmi's age and Niti's age will be 7 : 9. Two years ago, the respective ratio between Simmi's age and Niti's age was 1 : 3. If Abhay is 4 years older to his sister Niti, what is Abhay's present age? (in years)

[CET—Maharashtra (MBA) Exam, 2016]

- (a) 8 (b) 4
(c) 16 (d) 12
(e) 20

ANSWERS

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|---------|---------|---------|---------|---------|---------|---------|---------|---------|---------|
| 1. (c) | 2. (a) | 3. (b) | 4. (b) | 5. (a) | 6. (b) | 7. (b) | 8. (a) | 9. (c) | 10. (e) |
| 11. (d) | 12. (b) | 13. (c) | 14. (b) | 15. (d) | 16. (c) | 17. (b) | 18. (b) | 19. (c) | 20. (a) |
| 21. (b) | 22. (a) | 23. (b) | 24. (d) | 25. (c) | 26. (b) | 27. (c) | 28. (b) | 29. (c) | 30. (c) |
| 31. (a) | 32. (a) | 33. (e) | 34. (e) | 35. (b) | 36. (b) | 37. (b) | 38. (c) | 39. (d) | 40. (d) |
| 41. (a) | 42. (d) | 43. (b) | 44. (d) | 45. (b) | 46. (a) | 47. (a) | 48. (b) | 49. (b) | 50. (b) |
| 51. (d) | 52. (d) | 53. (b) | 54. (a) | 55. (a) | 56. (a) | 57. (d) | | | |

SOLUTIONS

1. Let mother's age be $7x$ years. Then, daughter's age = x years. $\frac{7x-4}{x-4} = \frac{19}{1}$

$$\Rightarrow 7x - 4 = 19(x - 4)$$

$$\Rightarrow 19x - 7x = 76 - 4$$

$$\Rightarrow 12x = 72$$

$$\Rightarrow x = 6.$$

$$\text{Mother's age after 4 years} = (7x + 4)$$

$$= (7 \times 6 + 4) \text{ years} = 46 \text{ years.}$$

2. Let Nishi's age be $6x$ years. Then, Vinnee's age = $5x$ years.

$$\therefore \frac{6x+9}{5x+9} = \frac{9}{8} \Rightarrow 8(6x+9) = 9(5x+9)$$

$$\Rightarrow 48x - 45x = 81 - 72$$

$$\Rightarrow 3x = 9$$

$$\Rightarrow x = 3.$$

$$\text{Difference in their ages} = (6x - 5x) = x \text{ years} = 3 \text{ years.}$$

3. Let Amit's age be $2x$ years. Then, his father's age = $5x$ years.

$$\therefore \frac{2x+4}{5x+4} = \frac{5}{11} \Rightarrow 11(2x+4) = 5(5x+4)$$

$$\Rightarrow 22x + 44 = 25x + 20$$

$$\Rightarrow 3x = 24$$

$$\Rightarrow x = 8.$$

$$\text{Father's age 5 years ago} = (5x - 5) \text{ years}$$

$$= (5 \times 8 - 5) \text{ years} = 35 \text{ years.}$$

4. Let father's age be $17x$ years. Then, son's age = $7x$ years.

$$\frac{17x-6}{7x-6} = \frac{3}{1} \Rightarrow 3(7x-6) = 17x-6$$

$$\Rightarrow 21x - 18 = 17x - 6$$

$$\Rightarrow 4x = 12$$

$$\Rightarrow x = 3.$$

$$\therefore \text{Father's present age} = 17x \text{ years}$$

$$= (17 \times 3) \text{ years} = 51 \text{ years.}$$

5. Let Shakti's age be $8x$ years. Then, Kanti's age = $7x$ years.

$$\therefore \frac{8x+10}{7x+10} = \frac{13}{12} \Rightarrow 12(8x+10) = 13(7x+10)$$

$$\Rightarrow 96x + 120 = 91x + 130$$

$$\Rightarrow 5x = 10$$

$$\Rightarrow x = 2.$$

$$\text{Difference between their ages} = (8x - 7x) \text{ years}$$

$$= x \text{ years} = 2 \text{ years.}$$

$$6. \text{ A's age} = \left(44 \times \frac{6}{11}\right) \text{ years} = 24 \text{ years and B's age}$$

$$= (44 - 24) \text{ years} = 20 \text{ years.}$$

$$\text{Ratio of their ages after 8 years} = \frac{(24+8)}{(20+8)} = \frac{32}{28} = \frac{8}{7} = 8:7.$$

7. Let Farah's age 8 years ago be x years. Then, her present age = $(x + 8)$ years.